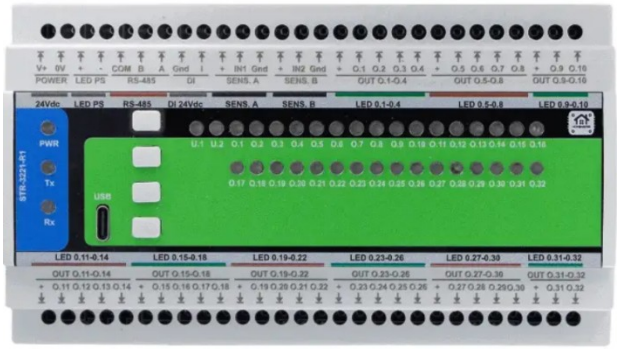
Datasheet

STR-3221-R1 Stair LED Controller Module



Part No.: STR-3221-R1

Manufacturer: ISYSTEMS AUTOMATION S.R.L.

Hardware Version: V1.0

Architecture: Modular (MCU Board + Field Board)

Controller: RP2350A

Functional Overview

The STR-3221-R1 is a DIN-rail mounted multi-channel LED and I/O control module for 12–24 V DC systems. It provides 32 low-side MOSFET outputs for LED loads, digital inputs, local user interface, RS-485 Modbus RTU communication, and USB-C configuration interface.

Board	Function
MCU Board	RP2350A controller, RS-485 communication, USB-C interface, power conversion
Field Board	32 MOSFET outputs, Digital inputs, protection and filtering

The module operates as a Modbus RTU slave over RS-485 and supports configuration and firmware update via USB-C.

Technical Specifications

Specification	Details
Microcontroller	RP2350A dual-core microcontroller
Storage	External QSPI Flash (32 Mbit)
Power Input	24 V DC nominal
Input Protection	1 A fuse (F8), TVS surge suppression, EMI filtering
Main Logic Supply	Buck regulator 24 V → 5 V (~4.96 V nominal), 3.3 V LDO regulator
LEDs supply input	External 12–24 V DC (independent from logic supply)
Document revision	DS-STR-3221-R1 Rev. B · 2026-09 · Hardware R1 (V1.0)
LED Outputs	32 × low-side MOSFET outputs (O1–O32), driven by 4 × TLC59208F (8 channels per chip). The TLC chips drive the main outputs, not the status LEDs.
LED Output Voltage	External 12–24 V DC supply
Maximum Total LED Current	18 A per module, shared across all channels in use
Maximum current per channel	1.5 A per channel
Output Structure	Open-drain, grouped by VCC rails, each channel with gate resistor + RC + ferrite
Digital Inputs	1 × module-wetted dry-contact input (ISO1212 input receiver, supplied from the module; not isolated)
Isolation	None — DI, IN1/IN2, outputs and RS-485 all share the module ground
Presence sensor inputs (SENS.A / SENS.B)	2 × presence inputs IN1 / IN2 (SFH6156), supplied from the module's SENS.A / SENS.B +5 V rails; not isolated
User Interface	4 buttons; 2 assignable status LEDs; power, RX and TX indicators; 32 channel-state indicators
Local control	Buttons SW1–SW4 and the three inputs act on the outputs; a local action holds priority until released.
Bus failsafe	Per-channel Hold / Off / Level on loss of the master; timeout 0 = off. Heat uses frost protection instead.
Sequencing	Stair sequence and heating-zone cycle run on the module.
Heating demand	Aggregate demand flag and open-zone count for the controller’s pump / boiler relay.
RS-485	Half-duplex Modbus RTU, not galvanically isolated; surge protection and fail-safe biasing. Factory default address 3, 19200 baud, 8N1.
USB	USB-C, 5 V logic, ESD protected
Modbus defaults	Address 3, 19200 baud, 8N1
Typical Power Consumption	0.2–0.5 W

Typical module power consumption: 0.2–0.5 W (logic only).

Maximum module power consumption: approximately 1W.

Top Terminal Block (1–28)

Front view – terminals numbered from left to right.

Term. No.	Marking	Signal	Description
1	V+	POWER	24 V DC input
2	0V	POWER	24 V DC return
3	+	LED_PS	LED PSU positive
4	-	LED_PS	LED PSU negative
5	COM	RS485	RS-485 signal reference
6	B	RS485	RS-485 data -
7	A	RS485	RS-485 data +
8	Gnd	DI	Discrete input return — dry contact only
9	I	DI	Discrete input — module-wetted 24 V, no external voltage
10	+	SENS_A	SENS.A group +5 V sensor supply
11	IN1	SENS_A	Presence input A — 5 V sourcing, ~4.3 mA, max 6.8 V
12	Gnd	SENS_A	SENS.A group return
13	+	SENS_B	SENS.B group +5 V sensor supply
14	IN2	SENS_B	Presence input B — 5 V sourcing, ~4.3 mA, max 6.8 V
15	Gnd	SENS_B	SENS.B group return
16	+	OUT_1_4	Group rail for O.1-O.4
17	O.1	OUT_1_4	Output 1
18	O.2	OUT_1_4	Output 2
19	O.3	OUT_1_4	Output 3
20	O.4	OUT_1_4	Output 4
21	+	OUT_5_8	Group rail for O.5-O.8
22	O.5	OUT_5_8	Output 5
23	O.6	OUT_5_8	Output 6
24	O.7	OUT_5_8	Output 7
25	O.8	OUT_5_8	Output 8
26	+	OUT_9_10	Group rail for O.9-O.10
27	O.9	OUT_9_10	Output 9
28	O.10	OUT_9_10	Output 10

Bottom Terminal Block (1–28)

Front view – terminals numbered from left to right.

Pos	Marking	Signal	Description
1	+	OUT_11_14	Group rail for O.11-O.14
2	O.11	OUT_11_14	Output 11
3	O.12	OUT_11_14	Output 12
4	O.13	OUT_11_14	Output 13
5	O.14	OUT_11_14	Output 14
6	+	OUT_15_18	Group rail for O.15-O.18
7	O.15	OUT_15_18	Output 15
8	O.16	OUT_15_18	Output 16
9	O.17	OUT_15_18	Output 17
10	O.18	OUT_15_18	Output 18
11	+	OUT_19_22	Group rail for O.19-O.22
12	O.19	OUT_19_22	Output 19
13	O.20	OUT_19_22	Output 20
14	O.21	OUT_19_22	Output 21
15	O.22	OUT_19_22	Output 22
16	+	OUT_23_26	Group rail for O.23-O.26
17	O.23	OUT_23_26	Output 23
18	O.24	OUT_23_26	Output 24
19	O.25	OUT_23_26	Output 25
20	O.26	OUT_23_26	Output 26
21	+	OUT_27_30	Group rail for O.27-O.30
22	O.27	OUT_27_30	Output 27
23	O.28	OUT_27_30	Output 28
24	O.29	OUT_27_30	Output 29
25	O.30	OUT_27_30	Output 30
26	+	OUT_31_32	Group rail for O.31-O.32
27	O.31	OUT_31_32	Output 31
28	O.32	OUT_31_32	Output 32

Installation, Environmental & Mechanical

Specification	Details
Terminal Type	Pluggable screw terminal blocks, 5.08 mm pitch
Wire Cross Section	0.2–2.5 mm^2 (AWG 24–12)
Tightening Torque	0.4–0.6 Nm

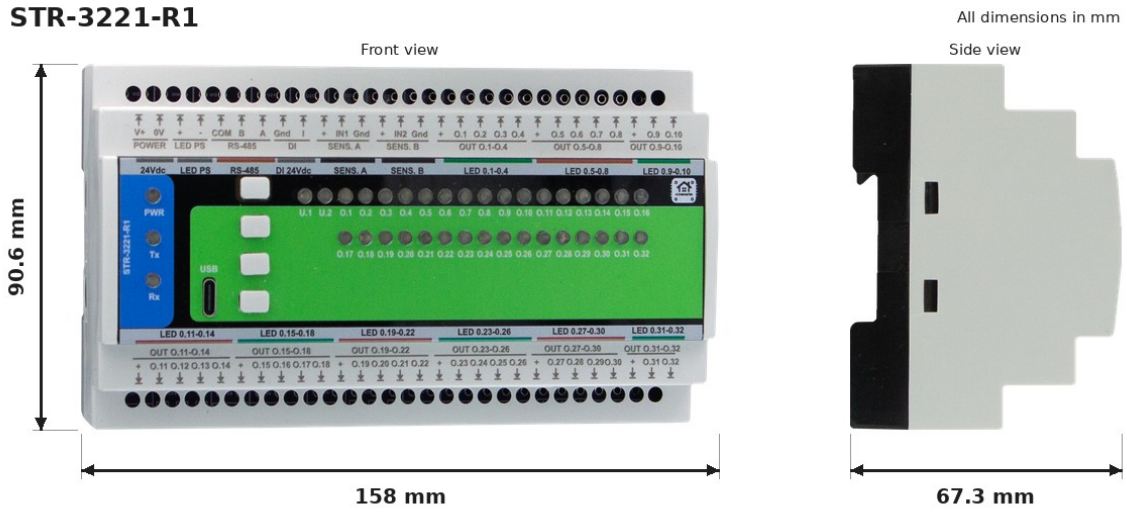
Specification	Details
Operating temperature	0 °C to +40 °C
Storage temperature	−10 °C to +55 °C

Relative humidity	0–90 % RH, non-condensing
Ingress protection	IP20 (inside cabinet only)
Installation	Indoor control cabinet only; not for outdoor or exposed installation
Maximum altitude	2000 m
Pollution degree	2

Specification	Details
Dimensions	158 × 90.6 × 67.3 mm (L × W × H)
DIN width	9 modules (≈ 158 mm)
Mounting	35 mm DIN rail
Enclosure	PC/ABS industrial enclosure
Terminal type	Pluggable screw terminal blocks, 5.08 mm pitch
Wire cross-section	0.2–2.5 mm² (AWG 24–12)
Tightening torque	0.4–0.6 Nm
Net weight	150 g
Gross weight	250 g
Pack size	230 × 140 × 87 mm (L × W × H)
Mechanical drawing	Mechanical drawing: front and side view, dimensions in mm
Notes	All dimensions shown in millimeters unless stated otherwise

Install only inside a control cabinet with ventilation; the cabinet must include a protective front plate covering all module connection terminals and a closing protective door; not for outdoor or exposed installation.

All wiring terminals must be protected against accidental contact by an insulating front plate, wiring duct, or terminal cover. Exposed live terminals are not permitted.



Cable Recommendations & Shield Grounding

This section applies to RS-485. Use shielded, twisted constructions and bond shields correctly to reduce EMI and ground-loop issues.

General Routing Rules

- Route low-level signal cables separately from power wiring.
- If crossing power cables is unavoidable, cross at 90°.
- Keep cable runs as short as practical.
- Avoid parallel runs with high-current conductors.

RS-485 Cable

- Twisted pair recommended (120 Ω characteristic impedance).
- Shielded cable recommended in industrial environments.
- Use one twisted pair for A/B.
- Run COM to every node. Required, not optional: the RS-485 port is not galvanically isolated.

Examples: J-Y(ST)Y 2×2×0.5 mm^2 or LI2YCY PiMF 2×2×0.50

Shield Grounding

- Bond cable shield(s) to cabinet PE/EMC ground at one end (PLC side recommended).
- Do not connect shields directly to signal terminals (A/B/COM).

Compliance & Certifications

The STR-3221-R1 module is CE marked and designed to comply with applicable European Union directives. The manufacturer maintains technical documentation and a signed EU Declaration of Conformity (DoC).

EMC 2014/30/EU · RoHS 2011/65/EU

Standards: EN 61000-6-1 · EN 61000-6-3 · EN 62368-1 · EN IEC 63000

Safety Notice

The power input and signal terminals are SELV circuits.

Installation by qualified personnel only.

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